

and 6 clocks per machine cycle modes, which are selectable via software. The Flash program memory supports both parallel programming and serial in-system programming (ISP).

Microcontroller IC10 is interfaced to the GSM module using MAX232 driver. The GSM module is a specialised type of modem that accepts a SIM card and operates on the subscriber's mobile number over a network, just like a cellular phone. Basically, SIM300 is a triband GSM/GPRS engine that works on EGSM 900MHz, DCS 1800MHz and PCS 1900MHz frequencies. GSM modem is RS232-logic-level compatible. Here MAX232 is used to convert TTL into RS232 logic level and vice versa.

Combination of resistor R7 and capacitor C4 provides the power-on reset for microcontroller IC10. Switch S2 is used for manual reset. An 11.0592MHz crystal (X_{TAL2}) along with two 33pF capacitors provides the basic clock frequency. Microcontroller IC10 receives a frame corresponding to every new valid message received by the GSM module. This frame is further processed and the message extracted out of it. Thereafter, an acknowledgment is sent through port pin 1.0 to the display section indicating that a new message will be transmitted.

Microcontroller IC9 receives this signal and immediately jumps to the receive module in the program and the message data is serially received. TXD and RXD lines of IC9 are connected to RXD and TXD pins of IC10.

Microcontroller IC9 controls the display unit. Once the complete message is received, it processes the data and serially sends it to the shift registers to drive the rows. The shift registers have a serial input (DS) that receives the message data through port pin P1.7 of IC9.

The data in each register is transferred to the storage register on a positive-going transition of the STCP input received from port pin P3.6 of the microcontroller IC9. Data is shifted on the positive-going transitions of the SHCP input received from port pin P3.7. Serial data output Q7S is used

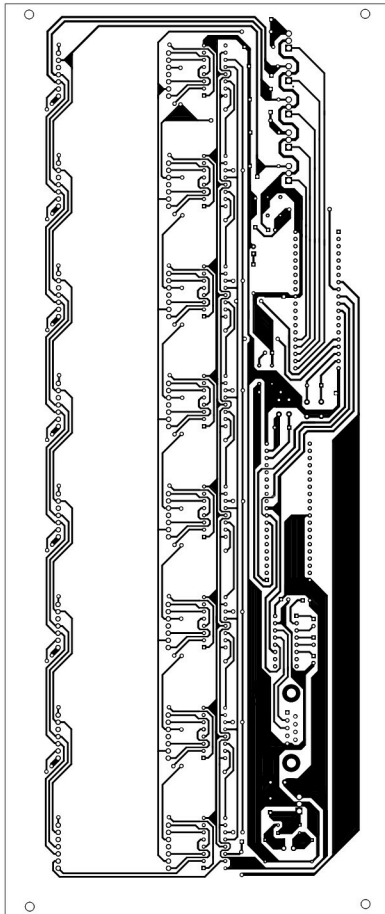


Fig. 4: Single-side PCB for the moving-message display (figure scaled down to 74 per cent)